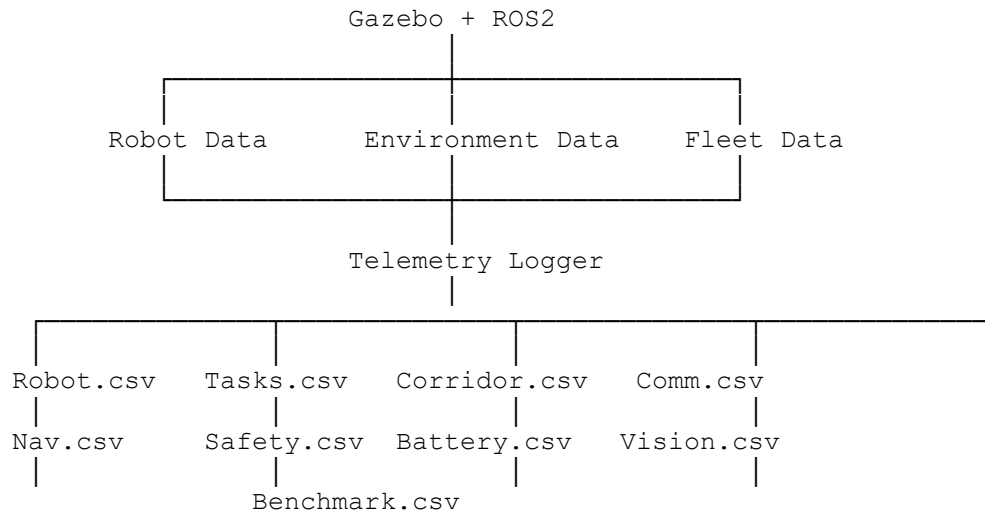


Master Data Collection Architecture



CSV 1 — robot_state.csv ★ ★ ★ ★ ★

This is the most important dataset.

One row every **100 ms** (10 Hz).

Column	Meaning
timestamp	Current simulation time
run_id	Simulation number
robot_id	Robot identifier
x	Position
y	Position
yaw	Orientation
linear_velocity	Current speed
angular_velocity	Turning speed
battery	Battery percentage
current_task	Current task ID
robot_state	Moving / Waiting / Charging
goal_x	Goal position
goal_y	Goal position
remaining_distance	Remaining path
planner_state	Planning/Executing

Column	Meaning
corridor_id	Current corridor
junction_id	Current junction
queue_length	Robots ahead
nearby_robot_count	Robots within 3 m

Example row

timestamp	run_id	robot_id	x	y	battery	task	state	corridor	queue
25.3	12	R2	6.5	3.2	82	T17	MOVING	C3	2

CSV 2 — task_log.csv ★ ★ ★ ★ ★

One row per completed task.

Column
task_id
run_id
pickup_location
drop_location
priority
creation_time
assignment_time
start_time
completion_time
assigned_robot
travel_distance
travel_time
waiting_time
success
failure_reason

Example

task	robot	pickup	drop	travel_time
T15	R2	Shelf4	Dock1	31 sec

CSV 3 — corridor_log.csv ★ ★ ★ ★ ★

This becomes your ML dataset.

One row **every** corridor traversal.

Column
run_id
robot_id
corridor_id
corridor_length
entry_time
exit_time
travel_time
queue_length
robots_inside
reservation_wait
blocked
replanned
battery
remaining_distance

Example

corridor	queue	robots	battery	travel_time
C12	3	4	81	18.2

🔗 **Target for ML**

travel_time

Everything else is the input.

CSV 4 — reservation_log.csv

Every reservation request.

Column

request_id
robot
corridor
request_time
grant_time
enter_time
exit_time
wait_time
expired
priority

Example

robot corridor wait

R3 C11 4 sec

Later ML predicts reservation delay.

CSV 5 — congestion_log.csv ★ ★ ★ ★ ★

One row every second.

Column

timestamp
corridor
robot_count
average_speed
queue_length
blocked
travel_delay
reservation_requests
density

Example

corridor robots queue delay

C8	6	4	12 sec
----	---	---	--------

This becomes the congestion prediction dataset.

CSV 6 — navigation_log.csv

One row per navigation request.

Column

robot
planner
start
goal
planned_distance
actual_distance
planning_time
travel_time
replans
collision_warning

Example

robot planner planned actual

R2	Smac	15.3	16.8
----	------	------	------

CSV 7 — battery_log.csv

Battery usage.

Column

timestamp
robot
battery
current
voltage

Column

charging
energy_used

Example

robot battery

R2 81%

CSV 8 — communication_log.csv

Important for decentralized systems.

Column

timestamp
sender
receiver
message_type
latency
packet_loss
retry

Example

sender receiver latency

R1 R2 34 ms

CSV 9 — safety_log.csv

Whenever safety triggers.

Column

robot
event
distance
speed
reaction_time

Column

stop_time

Example

robot event

R1 Emergency Stop

CSV 10 — benchmark.csv ★ ★ ★ ★ ★

One row per simulation.

Column

run_id

robots

tasks

makespan

average_wait

throughput

collisions

deadlocks

energy

success_rate

Example

run robots tasks makespan

25 5 50 480 sec

This is what you'll show in graphs.

CSV 11 — vision_annotations.csv

Links images to labels.

Column

image
robot
timestamp
class
bbox_x
bbox_y
bbox_w
bbox_h

Example

image	class
img125.jpg	Pallet

CSV 12 — rl_episode.csv ★ ★ ★ ★ ★

This is NOT a normal CSV for supervised learning, but it's useful for debugging and analyzing training.

episode	step	state_summary	action	reward	next_state_summary	done
42	1	R1@C3,Q=2	Assign T7→R2	+8	R2 moving	False
42	2	R2 entering C3	Wait	+2	Queue cleared	False
42	3	Task complete	Charge R2	+20	Battery charging	True

Internally, the RL algorithm stores richer tensors in a replay buffer, but a CSV like this is great for visualization and debugging.

Folder Structure

```
dataset/  
  
robot_state.csv  
  
task_log.csv  
  
corridor_log.csv  
  
navigation_log.csv
```

reservation_log.csv

congestion_log.csv

battery_log.csv

communication_log.csv

safety_log.csv

benchmark.csv

vision_annotations.csv

rl_episode.csv

images/

bags/